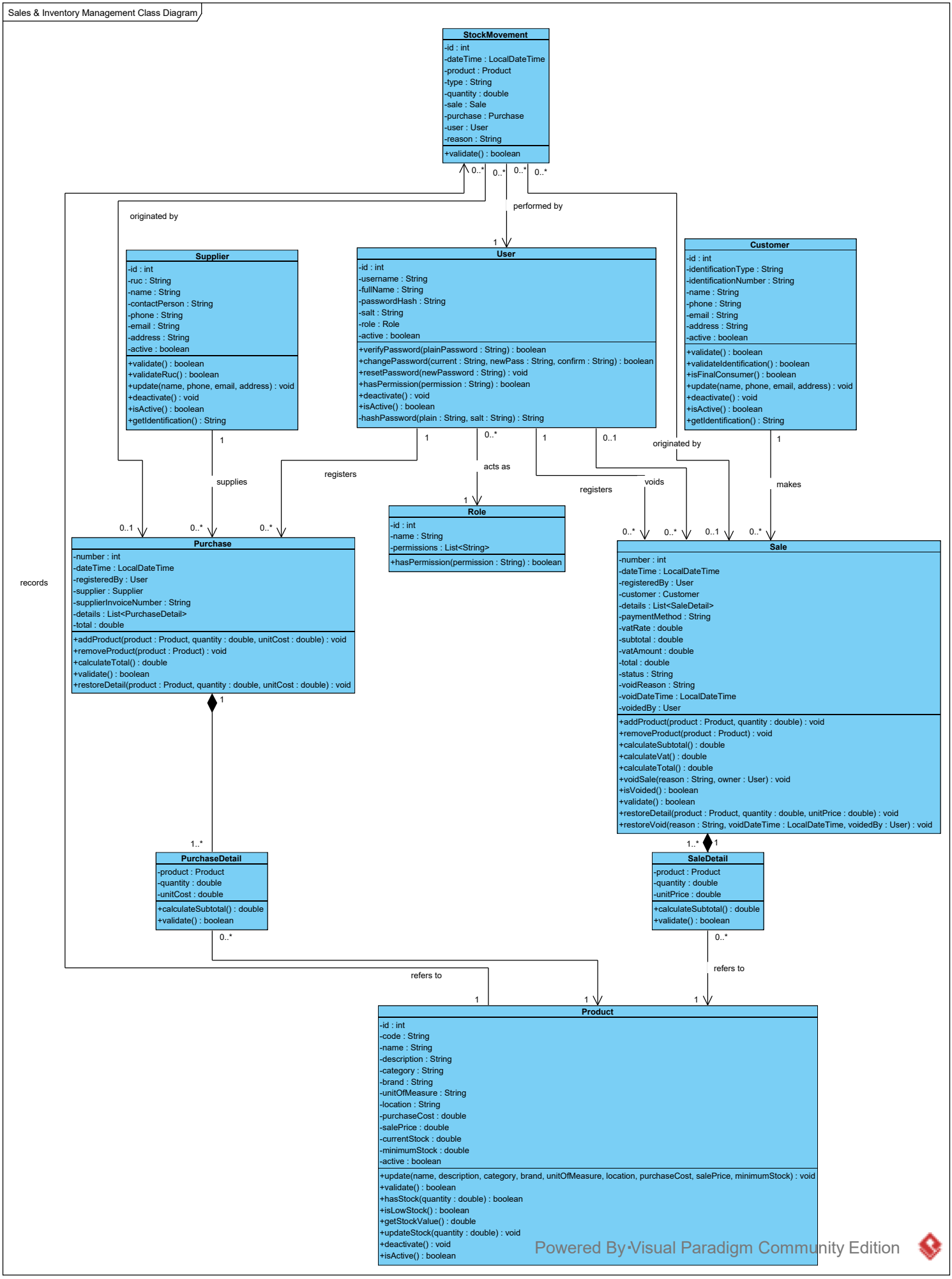
Sales & Inventory Management Class Diagram
StockMovement
-id : int
-dateTime : LocalDateTime
-product : Product
-type : String
-quantity : double
-sale : Sale
-purchase : Purchase
-user : User
-reason : String
+validate() : boolean
originated by
performed by
Supplier
-id : int
-ruc : String
-name : String
-contactPerson : String
-phone : String
-email : String
-address : String
-active : boolean
+validate() : boolean
+validateRuc() : boolean
+update(name, phone, email, address) : void
+deactivate() : void
+isActive() : boolean
+getIdentification() : String
User
-id : int
-username : String
-fullName : String
-passwordHash : String
-salt : String
-role : Role
-active : boolean
+verifyPassword(plainPassword : String) : boolean
+changePassword(current : String, newPass : String, confirm : String) : boolean
+resetPassword(newPassword : String) : void
+hasPermission(permission : String) : boolean
+deactivate() : void
+isActive() : boolean
-hashPassword(plain : String, salt : String) : String
Customer
-id : int
-identificationType : String
-identificationNumber : String
-name : String
-phone : String
-email : String
-address : String
-active : boolean
+validate() : boolean
+validateIdentification() : boolean
+isFinalConsumer() : boolean
+update(name, phone, email, address) : void
+deactivate() : void
+isActive() : boolean
+getIdentification() : String
supplies
registers
acts as
registers
voids
originated by
makes
records
Role
-id : int
-name : String
-permissions : List<String>
+hasPermission(permission : String) : boolean
Purchase
-number : int
-dateTime : LocalDateTime
-registeredBy : User
-supplier : Supplier
-supplierInvoiceNumber : String
-details : List<PurchaseDetail>
-total : double
+addProduct(product : Product, quantity : double, unitCost : double) : void
+removeProduct(product : Product) : void
+calculateTotal() : double
+validate() : boolean
+restoreDetail(product : Product, quantity : double, unitCost : double) : void
Sale
-number : int
-dateTime : LocalDateTime
-registeredBy : User
-customer : Customer
-details : List<SaleDetail>
-paymentMethod : String
-vatRate : double
-subtotal : double
-vatAmount : double
-total : double
-status : String
-voidReason : String
-voidDateTime : LocalDateTime
-voidedBy : User
+addProduct(product : Product, quantity : double) : void
+removeProduct(product : Product) : void
+calculateSubtotal() : double
+calculateVat() : double
+calculateTotal() : double
+voidSale(reason : String, owner : User) : void
+isVoided() : boolean
+validate() : boolean
+restoreDetail(product : Product, quantity : double, unitPrice : double) : void
+restoreVoid(reason : String, voidDateTime : LocalDateTime, voidedBy : User) : void
PurchaseDetail
-product : Product
-quantity : double
-unitCost : double
+calculateSubtotal() : double
+validate() : boolean
SaleDetail
-product : Product
-quantity : double
-unitPrice : double
+calculateSubtotal() : double
+validate() : boolean
refers to
refers to
Product
-id : int
-code : String
-name : String
-description : String
-category : String
-brand : String
-unitOfMeasure : String
-location : String
-purchaseCost : double
-salePrice : double
-currentStock : double
-minimumStock : double
-active : boolean
+update(name, description, category, brand, unitOfMeasure, location, purchaseCost, salePrice, minimumStock) : void
+validate() : boolean
+hasStock(quantity : double) : boolean
+isLowStock() : boolean
+getStockValue() : double
+updateStock(quantity : double) : void
+deactivate() : void
+isActive() : boolean
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